

1. Clinical Motivation

Ultrasound biometry underpins diagnosis across fetal, cardiac and vascular care. Manual landmark measurement is slow and operator-dependent. A unified model across all nine tasks would simplify deployment, but what limits accuracy: loss, resolution, or encoder capacity?

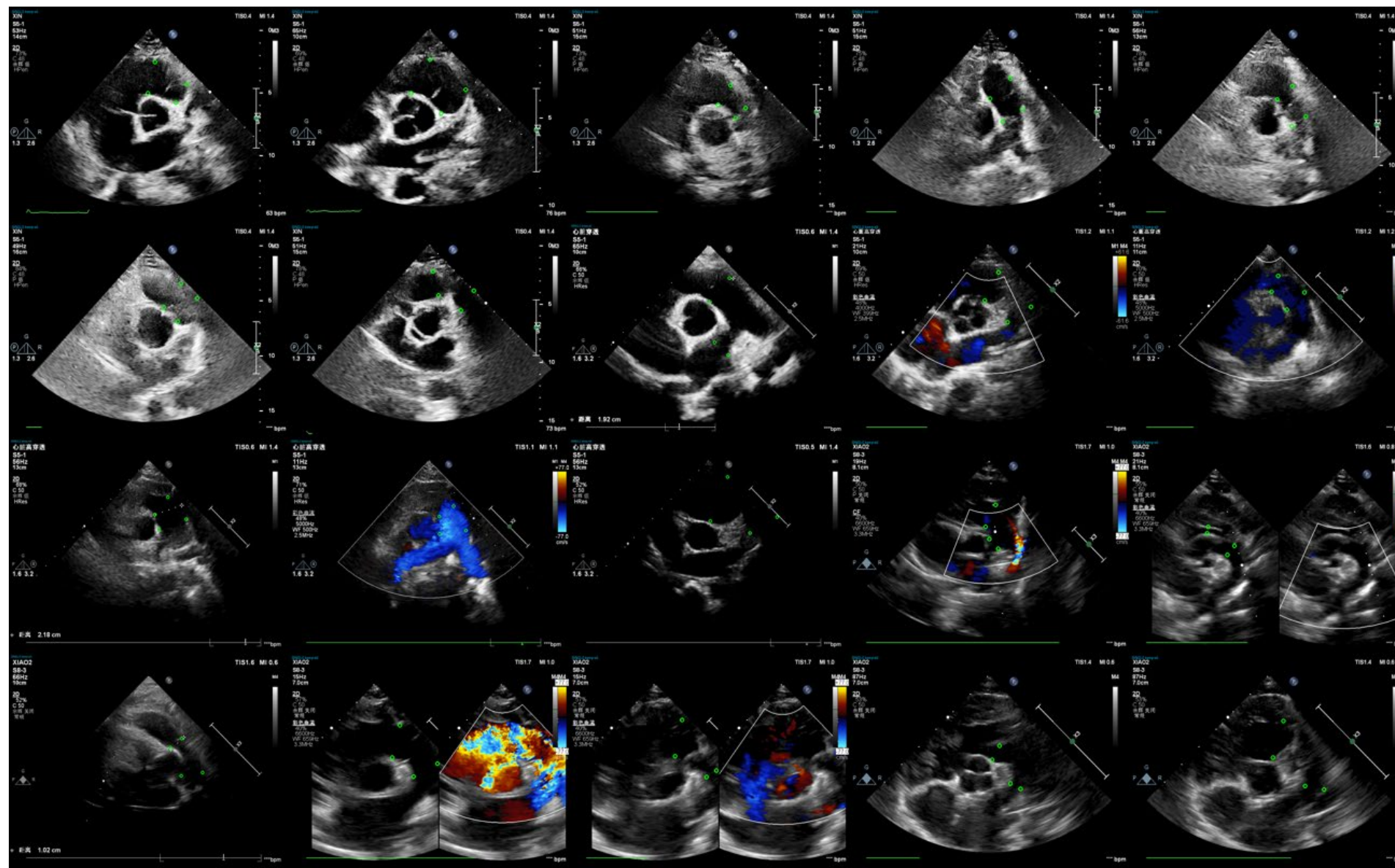
Aim: identify the binding constraint via a controlled ablation.

2. Dataset & Setup

Official FoundUS 2026 benchmark: nine ultrasound biometry tasks, 200K+ images (prenatal, intrapartum, cardiac, vascular).

Split per-task stratified 80/20 (seed 42)

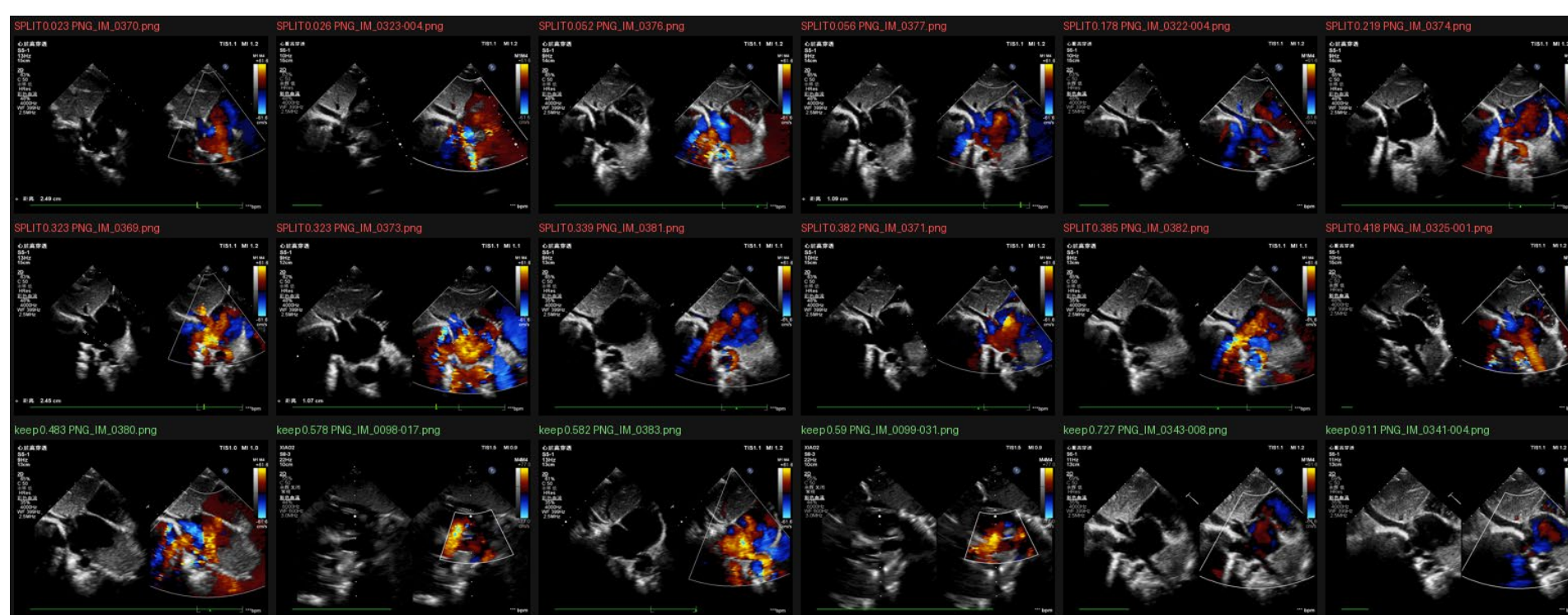
Metric Mean Radial Error (MRE), px



Annotated ultrasound (cardiac PSAX): expert landmarks per task.

3. Split-Screen Detection

Some cardiac acquisitions are dual-panel. A training-time detector crops the active panel for large gains.



Split-screen (red) vs. single-panel kept (green). IVC 40.2 to 2.76, PSAX 27.4 to 1.70 px (about 16x lower).

4. Key Contributions

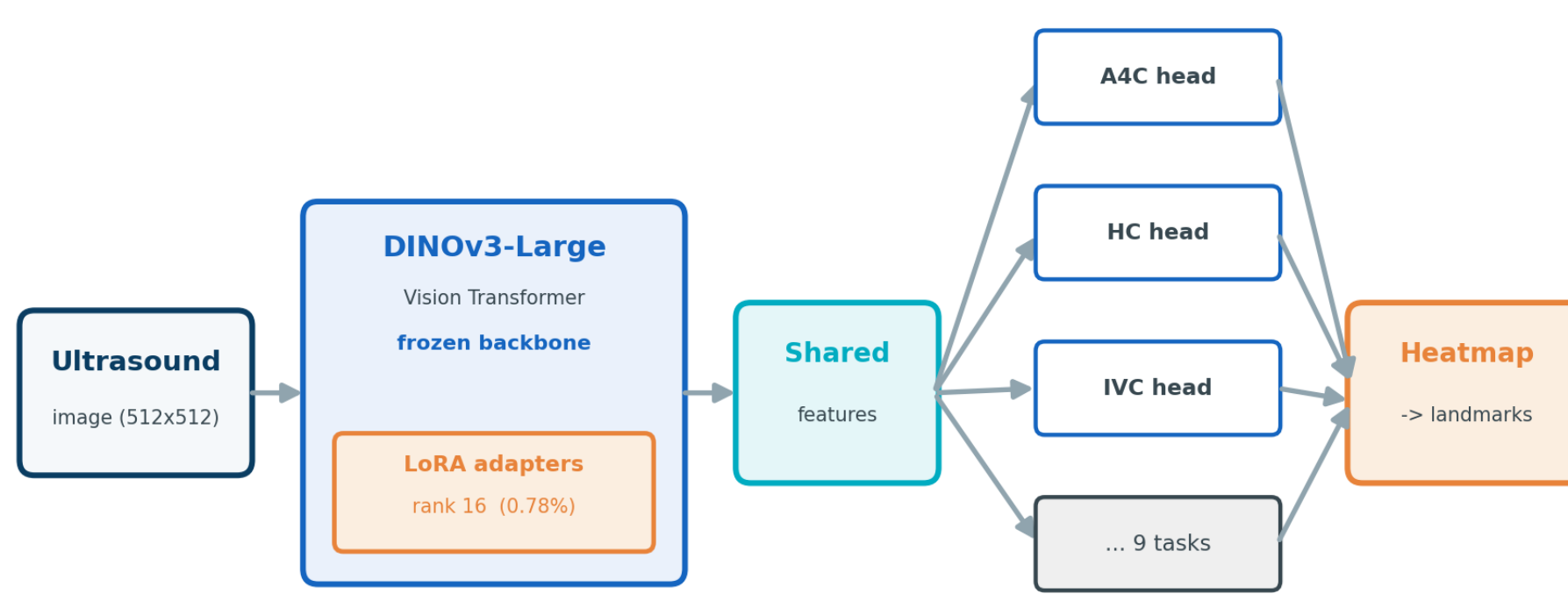
- Unified DINOv3 + LoRA model for all 9 tasks, training only 0.78% of parameters.
- Controlled ablation isolating encoder capacity as the binding constraint.
- Split-screen detection and acquisition-geometry covariate-shift analysis.

References

- [1] Oquab et al. DINOv2 / DINOv3: self-supervised vision features. 2024.
- [2] Hu et al. LoRA: Low-Rank Adaptation of Large Language Models. ICLR 2022.
- [3] FoundUS 2026: Foundation Model for Ultrasound Biometry, MICCAI 2026.
- [4] Deng et al. Baseline Method for the Foundation Model Challenge. 2026.

5. Method

A single frozen DINOv3-Large encoder is shared across all nine tasks and adapted with LoRA. Per-task heatmap heads decode landmark coordinates.

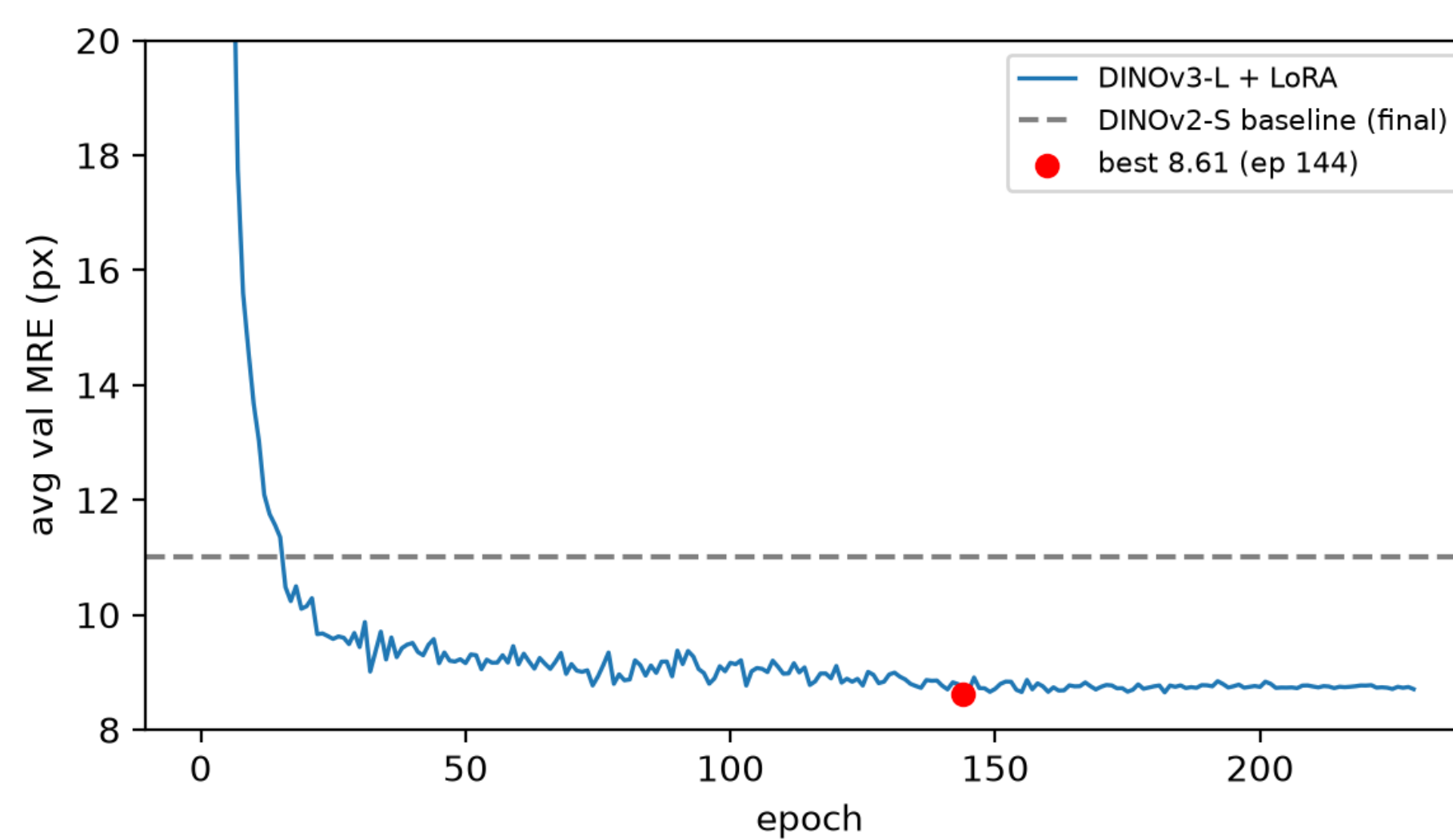


One frozen DINOv3 encoder + LoRA, shared across all nine tasks; lightweight per-task heatmap heads decode landmark coordinates.

Only LoRA adapters (2.39M of 305M = 0.78%) and per-task heads are trained; backbone frozen. Loss: MSE on Gaussian landmark heatmaps.

Controlled ablation: vary one factor at a time (encoder, loss, resolution), architecture fixed.

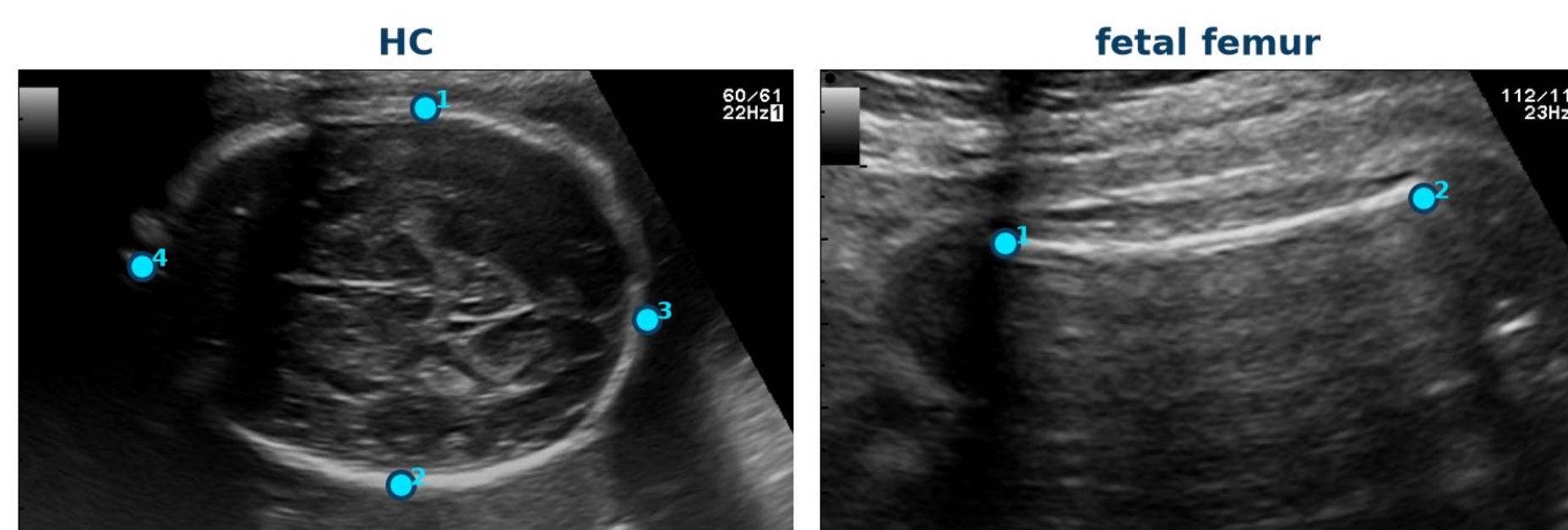
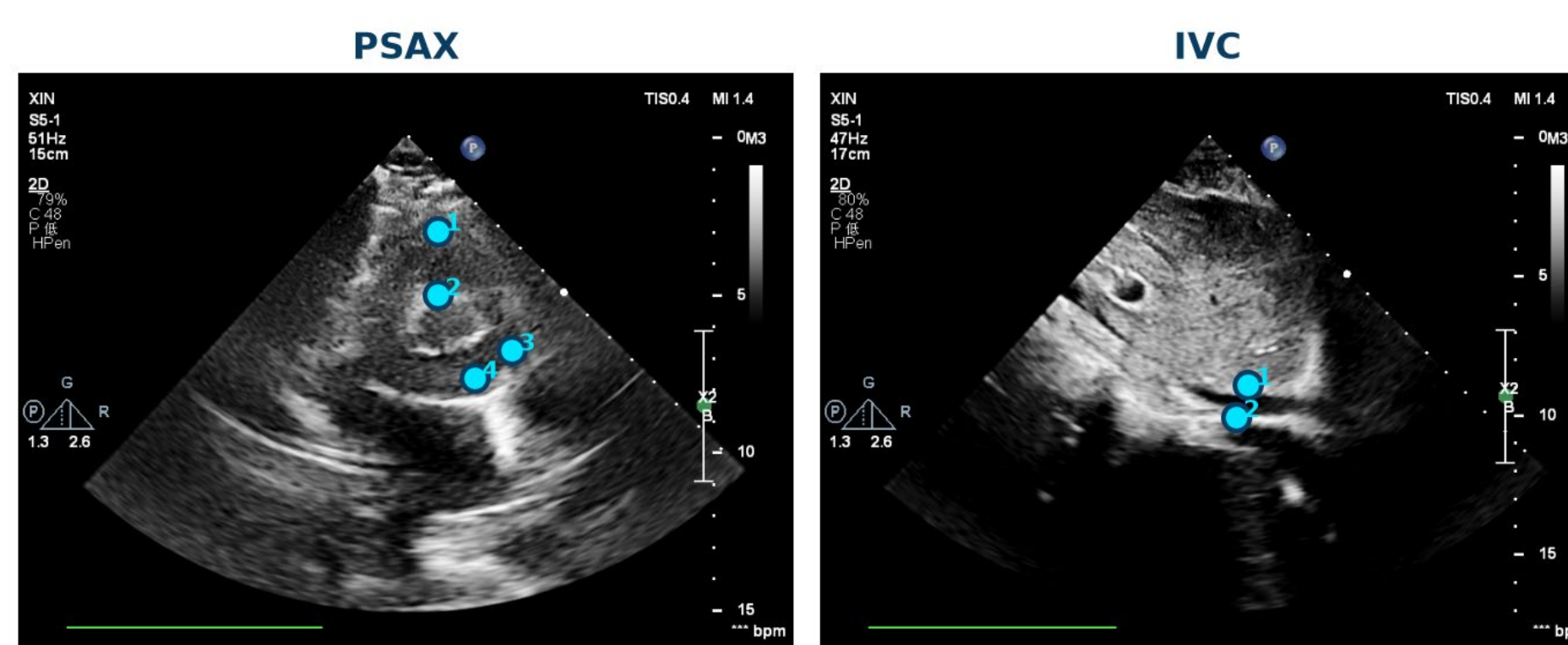
6. Training Convergence (Fig. 1)



DINOv3-L + LoRA (blue) falls well below the DINOv2-S baseline (dashed); best 8.61 px at epoch 144.

7. Qualitative: Predictions

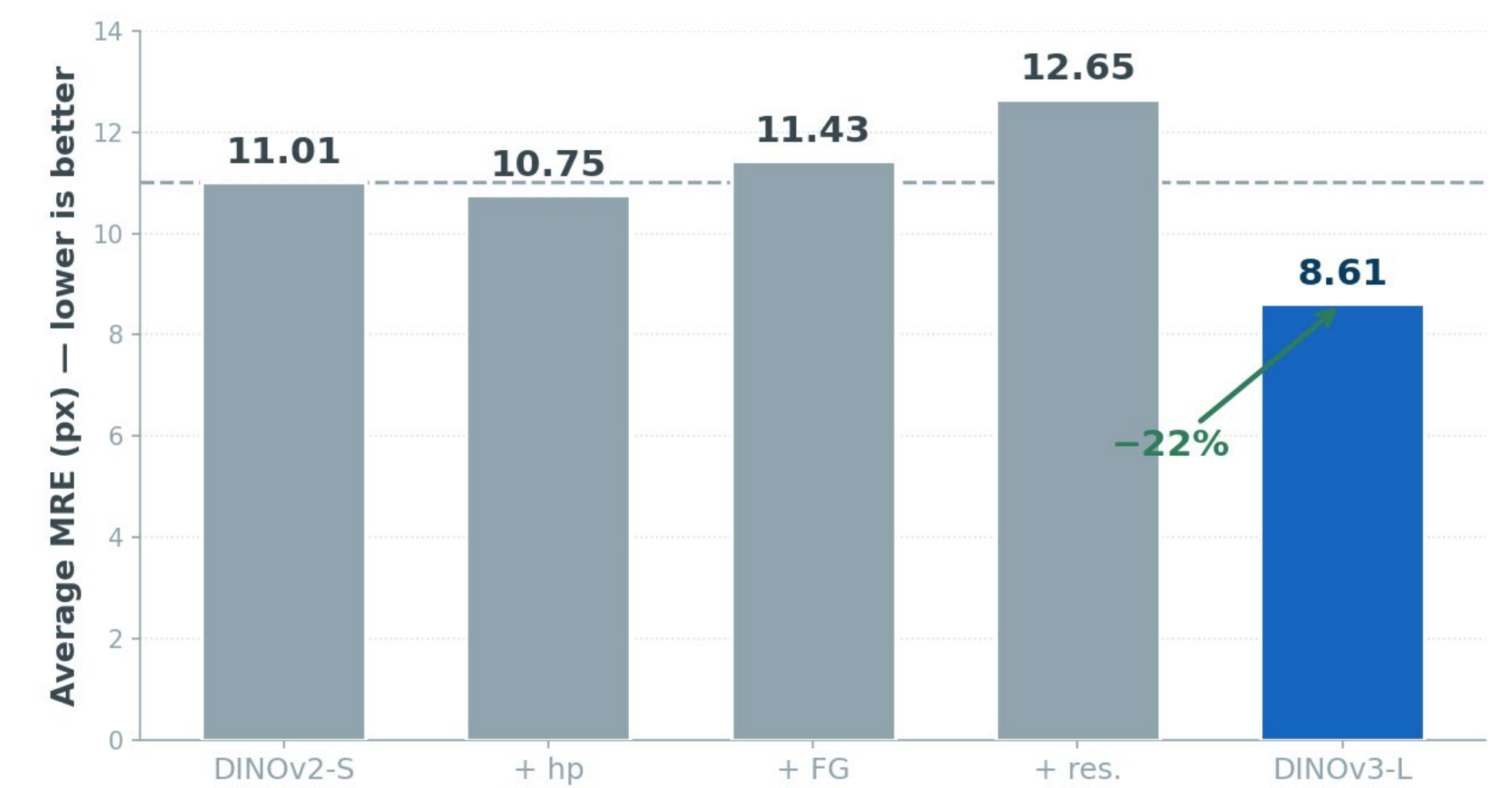
Model landmark predictions on held-out validation ultrasound



Limitations & Future Work

- Test geometries outside training range not covered.
- Single-model inference; no ensemble or TTA.
- Future: geometry-aware augmentation.
- Future: multi-centre training data.

8. Main Results



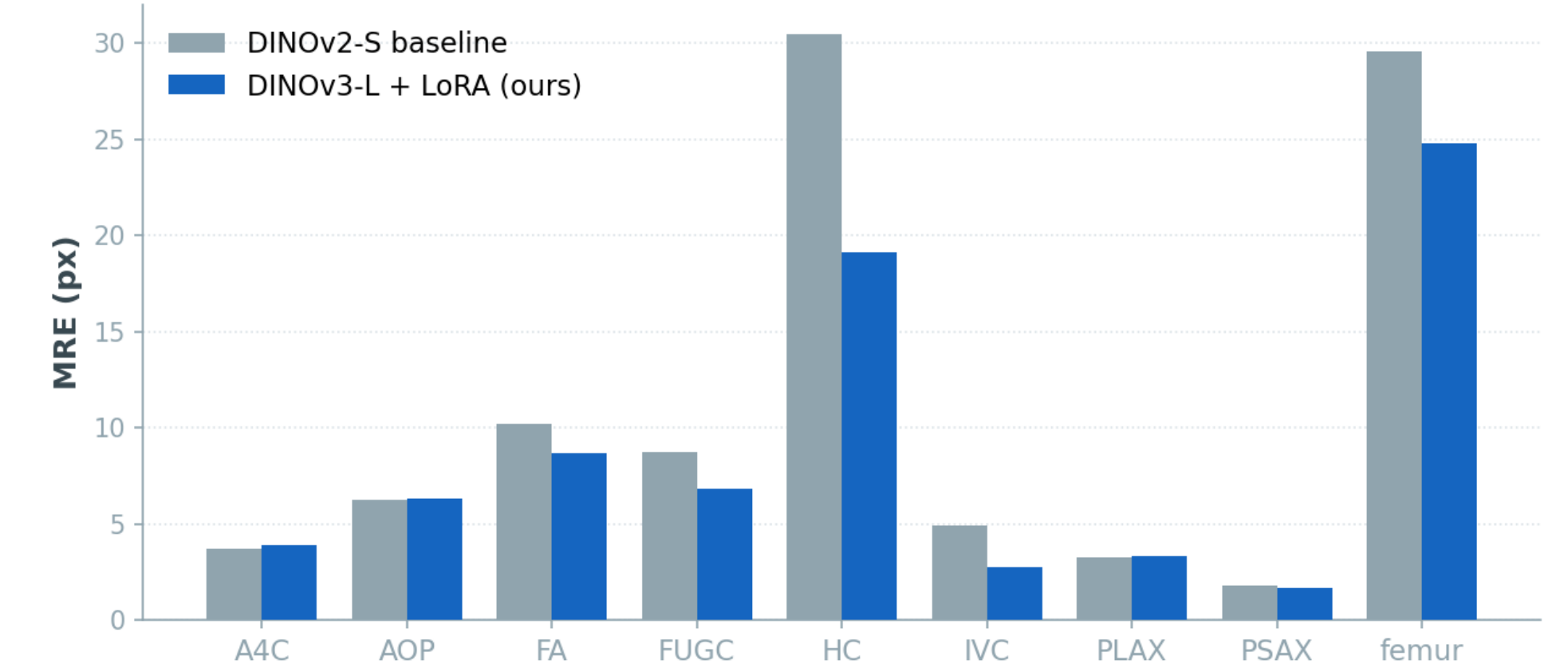
Best: encoder swap alone cuts avg MRE 11.01 to 8.61 px (22% lower), HC 30.5 to 19.1.

Per-task landmark error (Mean Radial Error, px — lower is better)

Encoder	Avg	A4C	AOP	FA	FUGC	HC	IVC	PLAX	PSAX	Femur
DINOv2-S (baseline)	11.01	3.74	6.24	10.25	8.77	30.48	4.93	3.26	1.78	29.60
DINOv3-L + LoRA (ours)	8.61	3.93	6.36	8.68	6.82	19.12	2.76	3.35	1.70	24.81

Controlled ablation — per-task MRE (px, lower is better). Only the encoder swap (row 7) beats baseline.

#Configurat	Avg	A4C	AOP	FA	FUGC	HC	IVC	PLAX	Femur
0 DINOv2	11.01	3.74	6.24	10.25	8.77	30.48	4.93	3.26	29.60
1 hp vari	10.75	3.97	6.69	10.27	8.24	29.48	4.87	3.58	27.86
2 hp vari	11.17	4.19	6.22	10.44	9.04	31.34	4.12	3.30	30.09
3 hp vari	11.64	4.77	7.07	9.54	8.69	31.89	5.94	3.70	31.30
4 + FG-w	11.43	4.46	6.24	10.77	9.19	30.11	7.01	3.43	29.89
5 + heatr	12.65	4.49	7.20	12.06	10.12	37.35	5.20	2.86	33.51
7 DINOv	8.61	3.93	6.36	8.68	6.82	19.12	2.76	3.35	24.81



Per-task MRE: gains concentrate on the hardest tasks (HC, fetal femur).

9. Robustness: Geometry Shift

Local 8.61 px vs. server 25.60 px. A distortion test isolates the cause:

Native aspect 1.48 (training) **10.5 px**

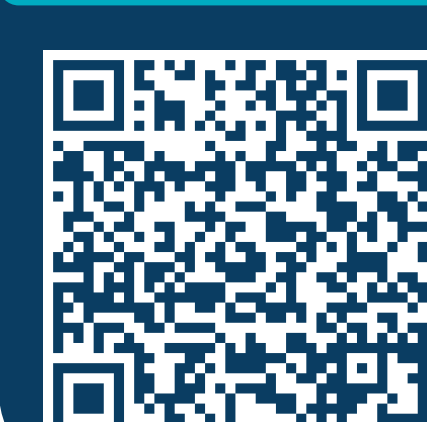
Stretched to 1.90 (server-like) **35.1 px**

Aspect-ratio change alone triples the error.

10. Conclusion

A single LoRA-adapted DINOv3 encoder addresses all nine ultrasound biometry tasks. A controlled ablation shows encoder capacity is the binding constraint (avg MRE 11.01 to 8.61); negative results rule out loss and resolution engineering. The residual local-to-server gap is acquisition-geometry covariate shift, the key direction for deployable multi-task ultrasound biometry.

Code & Contact



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